

Nicholas Mesa-Cucalon

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EXPERIENCE

- **REEP Equity** San Antonio, TX
AI Software Intern May 2025 – Present
 - **Prototype Development:** Built an end-to-end app with a web frontend and Flask/Python backend. Processed analyst CSV data and leveraged in-context learning with a fine tuned LLaMA-3.2 3B to generate summaries.
 - **User Testing and Validation:** Conducted A/B testing with clients comparing AI-generated and human-written summaries, demonstrating no significant preference and validating the prototype's effectiveness.
 - **Infrastructure Planning:** Scoping hardware and designing a ReactJS web interface with user authentication for an on-premises scalable solution. Planning Kubernetes orchestration to manage GPU workloads.
- **InfiniAI Lab CMU** Pittsburgh, PA
Student Researcher August 2024 – December 2024
 - **Research in Vision and Language Models:** Performed literature review on state-of-the-art token merging algorithms for vision models and key-value cache optimization methods for language models.
 - **Vision-Language Model Optimization:** Implemented a Heavy Hitter Oracle KV-Cache in the LLaVA vision-language model by modifying the LLaMA backbone. Matched vanilla LLaVA performance on TextVQA (61.24%) using only 25% of the original KV-Cache size, maintaining comparable inference speed.
- **Voaigne** Pittsburgh, PA
Robotics and Perception Intern May 2023 – August 2023
 - **Robot Calibration:** Led development and deployment of production-grade Delta Robot Calibration system with an 8-member team. Designed and documented a Hand-Eye Calibration solution for xArm7 robotic arms integrating Intel RealSense and Zivid cameras using OpenCV.
 - **Point Cloud Registration Research:** Conducted a literature review of 23 academic papers on Point Cloud Registration to synthesize key methods for development of the core product.

PROJECTS

- **Multimodal Meme Caption Generation:** Developed and implemented a model to use hybrid and late fusion of text and image data to generate captions for memes on the MemeCap dataset. Model achieved a BLEU-4 F1 Score of 9.0 and ROUGE-L F1 Score of 25.9, comparable to state of the art models.
- **“Denoising Diffusion Probabilistic Models” Implementation:** Wrote code implementing the forward process, reverse process and forward sampling for DDPM. Trained on cat images of the AFHQ dataset with a fixed compute of a single A100 GPU. Model was able to generate realistic cat images after training for 10,000 steps in 20 minutes.
- **“You need to Pay Better Attention” Paper Reimplementation:** Reimplemented “Optimized”, “Efficient” and “Super” Attention Layers in both PyTorch and Keras. Collaborated with original authors to replicate 78% test accuracy with the Super Layer on the IMDb Dataset.

EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA
B.S. in Electrical and Computer Engineering; Minor in Machine Learning Aug 2021 – May 2025
 - **Relevant Courses:** Computer Systems & Hardware-Software Interface, Distributed Systems, Machine Learning Systems, Bayesian Methods in Machine Learning, Computer Vision, Generative AI
- **Carnegie Mellon University** Pittsburgh, PA
M.S. in Electrical and Computer Engineering; Concentration in AI/ML Systems Expected: May 2026
 - **Relevant Courses:** Convex Optimization, Multimodal Machine Learning

SKILLS

- **Programming Languages:** Python, C, C++, Golang
- **Programming Frameworks:** PyTorch, OpenCV, Huggingface, Pandas, NumPy, Keras, Jupyter
- **Technologies:** Deep Learning, Computer Vision, Natural Language Processing, Weights and Biases, Git, Linux, AWS
- **Languages:** English (Native), Spanish (Native)